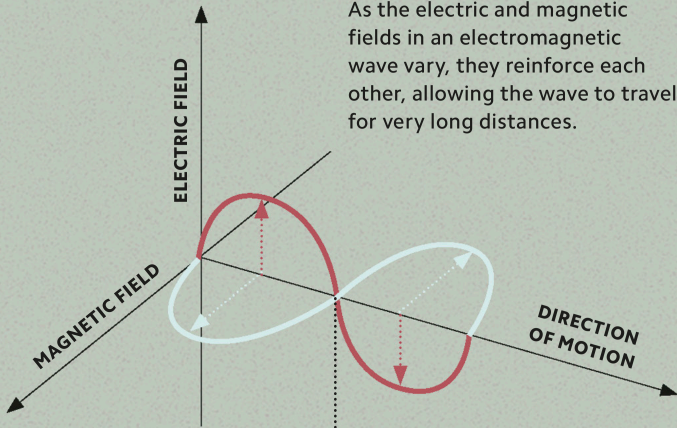
WHAT IS LIGHT?

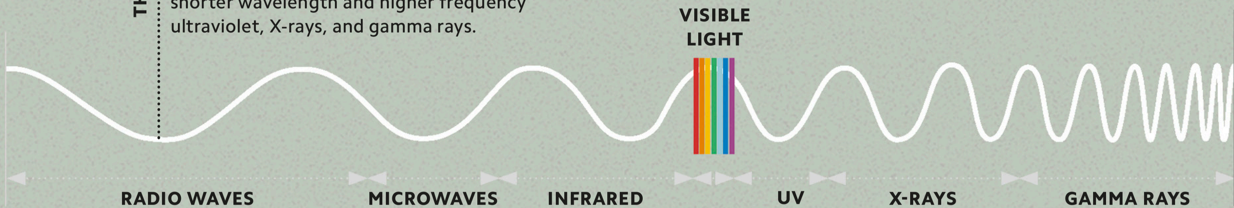
Particles with electric charge can exchange forces between each other by emitting electromagnetic radiation. These moving waves consist of oscillating electric and magnetic fields aligned at right angles so that changes in one reinforce the other. The properties and effects of these waves are determined by the frequency at which they repeat, giving rise to different forms of radiation such as X-rays, radio waves, and visible light.

Self-propagating wave
As the electric and magnetic fields in an electromagnetic wave vary, they reinforce each other, allowing the wave to travel for very long distances.



OSCILLATING FIELDS
Light consists of electrical and magnetic waves that are at right angles to each other and perpendicular to its direction of motion.

THE SPECTRUM
Scientists classify electromagnetic waves according to their wavelength and frequency, from long-wavelength, low-frequency radio waves, through microwaves, infrared, and visible light, to shorter wavelength and higher frequency ultraviolet, X-rays, and gamma rays.



ELECTROMAGNETIC SPECTRUM